

The speed of sound in a two-dimensional hard-disk gas: a divider-mode measurement against the equation of state

Christoph Haring Susanne Still

Abstract—We measure the adiabatic speed of sound of a hard-disk fluid from the eigenfrequency of a massive divider separating two compartments, the method of Román et al. (2002), over 35 packing fractions from $\eta = 0.0065$ to 0.76 and nine divider masses per density. The measurement is compared against the Kolafa–Rottner equation of state through the adiabatic mapping $c_s^2 = (k_B T/m)[Z + \eta Z' + Z^2]$ with no fitted parameter. For $\eta \leq 0.4$ the agreement is +1.0% on average and never worse than 1.7% at $N = 100$; extrapolating to infinite system size along a ladder of $N = 100$ –2500 brings it to $\lesssim 1\%$. We give the finite-size law, the estimator and its sensitivity, and a divider linewidth that recovers the Enskog transport coefficient of the same gas.

Index Terms—hard-disk fluid, speed of sound, event-driven molecular dynamics, equation of state, finite-size scaling

I. INTRODUCTION

Why the divider mode, why hard disks, what is being tested against what. State plainly that the EOS is an input, not a fit, so agreement is a test and disagreement is a measurement.

II. MODEL AND GEOMETRY

Hard disks of diameter $\sigma = 2r$ and mass m in a box $2L_0 \times H$, split by a divider of mass M and thickness t at the centre; N_s disks each side; $k_B T = m = \sigma = 1$; event-driven dynamics with the scientific backend (`--edmd-acc=0`). The divider is held for an equilibration period and then released; its position is recorded.

a) Effective length.: The centres of the disks in one compartment span r to $L_0 - t/2 - r$, so

$$L_{\text{eff}} = L_0 - 2r - t/2, \quad t = 0.05 \sigma. \quad (1)$$

Carrying t lowers every c_s by $(t/2)/(L_0 - 2r)$: 0.07% at $\eta = 0.10$ and 0.50% at $\eta = 0.65$.

III. THE ESTIMATOR

$$P(f) = \left| \sum_j (x_j - \bar{x}) e^{-2\pi i f t_j} \right|^2, \quad \nu = \arg \max_{f \geq \nu_{\text{pred}}/X} P(f), \quad (2)$$

The authors are with the Department of Information and Computer Sciences, University of Hawaii at Mānoa, Honolulu, HI 96822 USA (e-mail: charing@hawaii.edu; sstill@hawaii.edu).

per trajectory, then the mean over seeds, then c_s as the through-origin slope of $\bar{\nu}_M$ against $x_M = K(\alpha)/(2\pi L_{\text{eff}})$ with $\cot K = \alpha K$, $\alpha = M/2N_s m$. The one-sided edge exists because the divider carries a second, slow mode; the choice $X = 2.5$ is flat over $X \in [2.08, 6.25]$ at 200 periods.

a) Why the largest bin and not a resonance fit.: Both put the nine masses on one line to 0.27%, but the residual of the fit *leans* with mass, $-0.286 \pm 0.020\%$ per e-fold in α (14.4σ), while the largest bin is flat, $-0.030 \pm 0.029\%$ (1.0σ). The fit is the more precise estimator and the more biased one.

IV. RESULTS

Over $\eta \leq 0.4$ the measurement sits +1.04% above Kolafa–Rottner on average, with a maximum deviation of 1.68%. 7873 of 7875 trajectories were used; two were discarded by the health contract.

V. FINITE SIZE

Along $N = 100, 400, 900, 1600$ (and 2500 where available) at fixed η , $c_s(N)$ is fitted with $c_\infty + b/\sqrt{N}$; the two other forms (b/N and both together) bracket the systematic. At $\eta = 0.02$, $c_\infty = 1.4614 \pm 0.0102$ against 1.4725 from the EOS (-1.1σ); at $\eta = 0.05$, 1.5699 ± 0.0029 against 1.5666 ($+1.1\sigma$).

a) What the offset is not.: The wall pressure of the same finite box is 7–20% above the bulk EOS and rising with η , which would imply a +5 to +16% excess in c_s — five times what is observed. The wall value is a surface contact value; the mode samples the bulk. The finite-size law above is the measurement, not the wall pressure.

VI. THE DIVIDER LINEWIDTH GIVES A TRANSPORT COEFFICIENT

Linearising the piston equation of Malek Mansour et al. gives a damped oscillator with $Q = 1/\mu$, $\mu = \Gamma L_y \sqrt{2/(\hat{M}N)}$, $\hat{M} \propto M + 5\eta N/3$. The measured FWHM of the seed-averaged spectrum, with the bin width removed in quadrature, gives $\Gamma = 0.314 \pm 0.038$ at $\eta = 0.112$, against the Enskog value 0.331 — consistent within its 12% error. The $M = 50$ row ($\alpha = 0.5$) fails the $K \ll 1$ assumption and is excluded; the $M = 500$ row is 0.59 of prediction and is not explained.

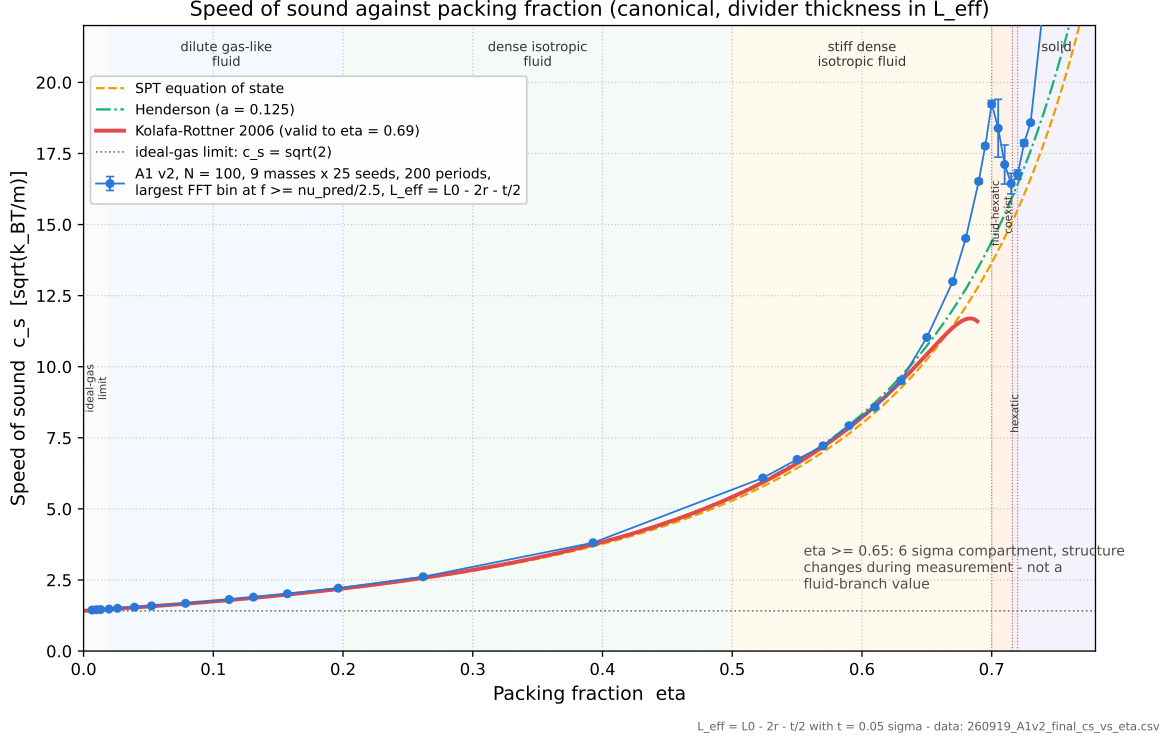


Fig. 1. $c_s(\eta)$ at $N = 100$ against three equations of state. Shading marks the approximate regimes; only 0.700/0.716/0.720 are physical transitions. Points beyond $\eta \approx 0.69$ lie in the coexistence/ordering region where no fluid reference exists; they are shaded as such and are not counted as a deviation from Kolafa-Rottner.

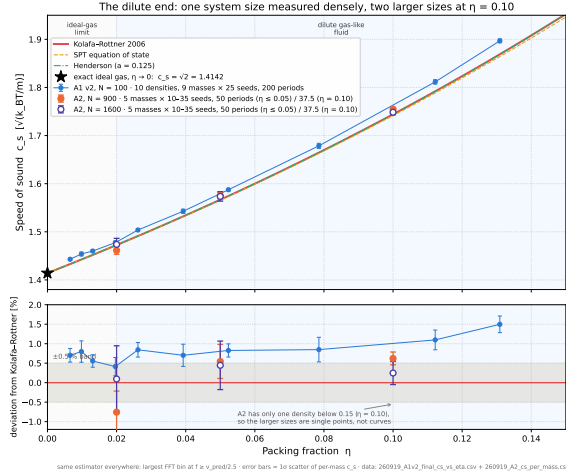


Fig. 2. The dilute end. The $N = 100$ curve sits above the EOS by 0.5–1.5%; the larger systems agree with it within their errors, so the offset is finite size.

VII. COMPARISON WITH ROMÁN ET AL. (2002)

Román et al. measured the same observable in the same box ($N_s = 50$ per side, $H = 10$) and report c_s lying above spt and Henderson by an amount that *grows* with density, from +1.3% at $\eta = 0.112$ to +12.8% at 0.524. That growth is the mapping, not the data. They convert Z to c_s with a constant $\gamma = 2$; the correct adiabatic relation for a two-dimensional monatomic hard-disk fluid is $c_s^2 = (k_B T/m)[Z + \eta Z' + Z^2]$.

Re-mapping their own numbers through it, against Kolafa-Rottner:

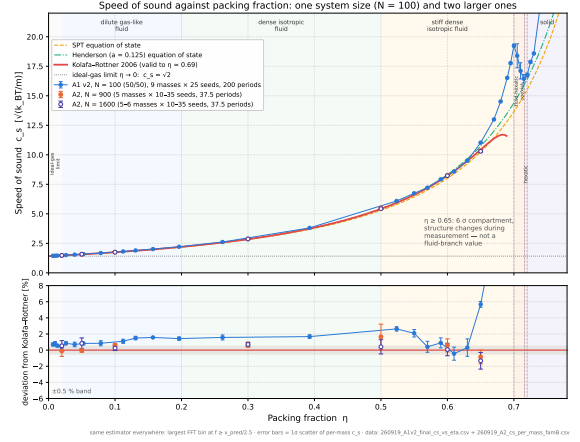


Fig. 3. $N = 100$ against $N = 900$ and 1600.

Their density-dependent excess becomes a *flat* +1.10% (spread +0.91 to +1.46%), and our $N = 100$ curve sits +1.04% above the same equation of state averaged over $\eta \leq 0.4$. Two independent implementations, twenty-four years apart, in the same box, with the same offset: that makes the +1% a property of the $N = 100$ divider apparatus rather than of either code. Their own size series (Table II, $N = 64 \rightarrow 4096$, $c_s = 3.81 \rightarrow 3.75$ extrapolating to 3.74) falls in the same direction as ours (§V).

As a transcription check, their spt and Henderson columns are reproduced to every printed digit by the equation-of-state code used here together with their $\gamma = 2$ mapping.

TABLE I
ROMÁN ET AL. (2002) TABLE I RE-MAPPED. η IS RECOMPUTED FROM THEIR L_0 .

L_0	η	c_s (theirs)	dev., $\gamma = 2$	dev., adiabatic
7.5	0.5236	5.99 ± 0.09	+10.6 %	+1.0 %
10	0.3927	3.78 ± 0.08	+5.5 %	+0.9 %
15	0.2618	2.61 ± 0.03	+3.3 %	+1.5 %
20	0.1963	2.20 ± 0.02	+1.9 %	+0.9 %
25	0.1571	2.01 ± 0.02	+1.9 %	+1.3 %
30	0.1309	1.89 ± 0.02	+1.5 %	+1.1 %
35	0.1122	1.81 ± 0.02	+1.3 %	+1.0 %

- $\eta = 0.65$ at larger N , and an $N \approx 1000$ sweep, as the final Paper 1 runs.
- The $M = 500$ linewidth row.

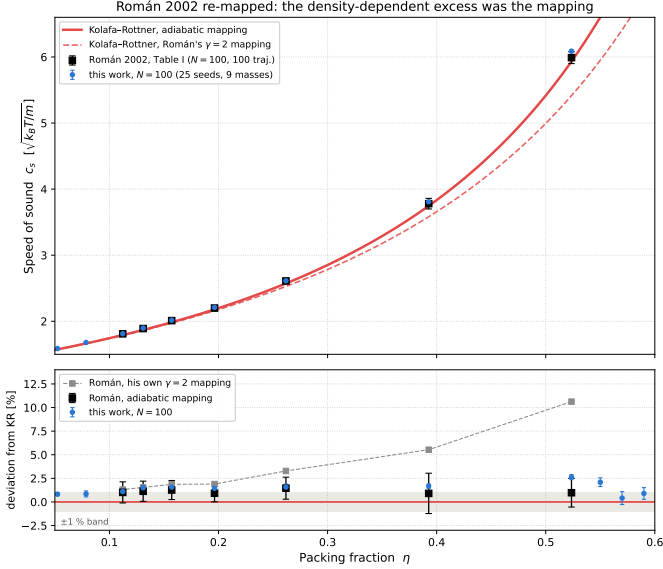


Fig. 4. Román et al. re-mapped. Solid red, Kolafa–Rottner through the adiabatic relation; dashed red, the same equation of state through their $\gamma = 2$ mapping; black squares, their Table I with their error bars; blue, this work at $N = 100$. Lower panel: deviation from Kolafa–Rottner, with their own mapping shown in grey for comparison.

The one point where the two measurements differ is $\eta \approx 0.52$: ours sits $+2.6\%$ and theirs $+1.0 \pm 1.5\%$, which is 1.1σ on their error bar alone, with 100 trajectories against our 25.

VIII. METHODS

Health contract, seeding, sampling, record length, discard rule.

IX. REPRODUCIBILITY

Every campaign directory carries `00_COMMAND.md` with the exact command and the estimator geometry line. The geometry constant and the estimator share one definition (`tests_20260913.WALL_T, l_eff`), and `assert_wall_thickness` refuses to analyse data whose recorded thickness disagrees.

OPEN

- $\eta > 0.69$: no fluid reference. Liu (2021) global EOS with hexatic and solid branches.